

Name:

Roll Number:

Section:

Q1 - 30 Marks

Criterion	Description	Marks
1. Decision Criteria	When to prefer Naïve / KMP / Rabin–Karp based on pattern length, alphabet, collisions.	4
2. Multi-pattern Preprocessing Optimization	Reducing preprocessing time (e.g., shared prefix tables, hashing windows).	3
3. Adaptive Runtime Switching	Detecting periodicity and switching algorithms dynamically.	3
4. Time Complexity Analysis	Case-wise complexity explanation.	3
5. Example Scenario	Demonstrates algorithm behavior.	2
6. Code Implementation	Correct, readable code.	10
7. Conceptual Question	Correct answer.	5

Q2 - 30 Marks

Criterion	Description	Marks
1. Divide Strategy	How documents are divided into blocks.	5
2. Merge Strategy	How partial LCS results from blocks are combined, ensuring cross-boundary correctness.	3
3. Time & Space Complexity Analysis	Case-wise complexity explanation.	3
4. Parallelization Discussion	How subproblems can run on separate processors and merge efficiently.	2
5. Example Scenario	Demonstrates algorithm behavior.	2
6. Code Implementation	Correct, readable code.	10
7. Conceptual Question	Correct answer.	5

Q3 - 35 Marks

Criterion	Description	Marks
1. Memory-Optimized Automaton	Conceptual design for memory reduction.	4
2. Divide & Conquer Construction	Suggest how splitting patterns and merging partial automata reduces preprocessing time.	4
3. Aho–Corasick Implementation	Correct implementation.	4
4. Efficiency Comparison	Compare with standard Aho–Corasick.	3
5. Pseudocode for Build + Merge	Provide high-level pseudocode.	3
6. Example Scenario	Demonstrates algorithm behavior.	2
7. Code Implementation	Correct, readable code.	10
8. Conceptual Question	Correct answer.	5

Q4 – 35 Marks

Criterion	Description	Marks
1. Divide Strategy	Splitting string into blocks + overlap windows.	4
2. Longest Repeated Substring	Efficient LRS inside and across blocks.	3
3. Merge Step Correctness	Ensures cross-boundary repeats aren't lost.	3
4. Compressed Representation	How repeated substrings map to final compressed format.	3
5. Time Complexity	Full asymptotic behavior.	3
6. Comparison with Suffix-array	Clear, accurate comparison.	2
7. Example Scenario	Demonstrates algorithm behavior.	2
8. Code Implementation	Correct, readable code.	10
9. Conceptual Question	Correct answer.	5